COURSE HANDOUT OUTLINE (CHO)

Mohan Babu University - Department of CSE (Data Science)
DATA SCIENCE DIGITAL LIBRARY
Official Course Curriculum & Syllabus Outline

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Course Overview & Objectives:

The Data Science Digital Library provides an industry-aligned, mathematically grounded, and computationally rigorous program. This Course Handout Outline (CHO) encapsulates the foundational through advanced concepts distributed across five primary curriculum modules: Foundations, Data Engineering & Wrangling, Data Visualization, Statistical Inference, and Predictive Time Series Analysis.

Students will acquire practical competencies in Python, scientific computing stacks, exploratory data analysis, visual storytelling, hypothesis testing, stochastic modeling, and high-performance predictive forecasting.

Curriculum Module 01: Introduction to Data Science

Scope & Description: Foundational paradigms, lifecycle stages, computational workflows, tool ecosystems, data engineering basics, and exploratory philosophy.

Key Sub-Topics & Detailed Syllabus Breakdown:

- [1.1] The Evolution of Data Science - Historical trajectory from classical statistics to big data and artificial intelligence.

- [1.2] The Data Science Lifecycle - Problem formulation, data acquisition, scrubbing, exploratory analysis, modeling, and operationalization.

- [1.3] Roles and Responsibilities - Differentiating Data Scientists, Data Engineers, Machine Learning Engineers, and BI Analysts.

- [1.4] Tools & Scientific Computing Stack - Architecture of Python, NumPy, Pandas, Jupyter, and cloud data platforms.

- [1.5] Ethics, Privacy & Governance - Data provenance, GDPR compliance, algorithmic bias, and ethical stewardship.

- [1.6] Data Engineering Basics - File formats (Parquet, CSV, Feather), database schemas, and ETL/ELT pipelines.

- [1.7] Industry Applications - Healthcare diagnostics, financial risk modeling, predictive maintenance, and e-commerce recommendations.

- [1.8] Traditional Programming vs Data Science - Rule-based heuristics versus data-driven probabilistic learning.

- [1.9] Environment Setup & Reproducibility - Virtual environments, Conda, Docker containers, and version control with Git.

- [1.10] Future Horizons in Data Science - Autonomous agents, generative AI synthesis, AutoML, and edge intelligence.

Curriculum Module 02: Data Extraction & Wrangling

Scope & Description: Data harvesting, Web Scraping, RESTful API consumption, schema normalisation, missing value imputation, and pipeline orchestration.

Key Sub-Topics & Detailed Syllabus Breakdown:

- [2.1] Introduction to Data Sources - Heterogeneous sources: relational databases, streaming feeds, unstructured web sources, and public repositories.

- [2.2] Web Scraping Foundations - DOM architecture, HTTP protocol, robots.txt compliance, and scraping legality.

- [2.3] HTML & CSS Selectors - Parsing tree navigation, XPath expressions, and targeting specific DOM nodes.

- [2.4] Python Requests & BeautifulSoup - Automated HTTP session management, response parsing, and HTML soup navigation.

- [2.5] Working with RESTful APIs - Endpoints, HTTP methods (GET, POST), headers, rate limits, and JSON payloads.

- [2.6] Authentication & Security - Bearer tokens, OAuth 2.0 handshakes, API keys, and credential encryption.

- [2.7] Parsing JSON, XML & Nested Data - Serialization, deserialization, schema flattening, and handling semi-structured hierarchies.

- [2.8] SQL Database Connections - SQLAlchemy, connection pooling, executing queries via Pandas, and bulk ingestions.

- [2.9] Pagination, Rate Limiting & Robustness - Exponential backoff, jitter, proxy rotation, and asynchronous extraction (aiohttp).

- [2.10] Wrangling Best Practices & Imputation - Anomaly detection, IQR filtering, KNN imputation, and data type coercion.

Curriculum Module 03: Data Visualization

Scope & Description: Visual perception hierarchies, color theory, exploratory plotting with Matplotlib and Seaborn, interactive dashboards, and storytelling.

Key Sub-Topics & Detailed Syllabus Breakdown:

- [3.1] Importance of Visualization & Perception - Graphical communication efficiency, Cleveland-McGill perceptual accuracy, and visual encoding.

- [3.2] Human Perception & Color Theory - Preattentive processing, Gestalt principles, color spaces, and accessible color palettes (Viridis).

- [3.3] Introduction to Matplotlib Architecture - Figure, Axes, Artists, canvas rendering pipeline, and object-oriented plotting API.

- [3.4] Statistical Plots with Seaborn - Box plots, violin plots, KDE distributions, pair plots, and categorical matrix visualization.

- [3.5] Interactive Dashboards & Controls - Plotly figures, Dash callbacks, tooltips, interactive slicing, and browser-based graphics.

- [3.6] Geospatial Visualization & Mapping - Coordinate reference systems (CRS), GeoPandas, Folium tile maps, and choropleth projections.

- [3.7] Time Series & Longitudinal Visuals - Line chart conventions, rolling averages, seasonality decompositions, and calendar heatmaps.

- [3.8] Visual Integrity & Chart Ethics - Lie factors, avoiding 3D distortion, truncated axes pitfalls, and ethical data representation.

- [3.9] Storytelling with Data - Editorial presentation, narrative arc, annotation callouts, and guiding audience attention.

- [3.10] Advanced Multi-Panel Techniques - GridSpec layouts, facet grids, custom style sheets, and publication-ready vector exports (SVG/PDF).

Curriculum Module 04: Statistical Thinking & Inference

Scope & Description: Probability theory, random variables, sampling distributions, Central Limit Theorem, hypothesis testing, ANOVA, and Bayesian reasoning.

Key Sub-Topics & Detailed Syllabus Breakdown:

- [4.1] Introduction to Probability Theory - Axiomatic probability, sample spaces, conditional probability, and Bayes' Theorem.

- [4.2] Descriptive Statistics & Exploratory Metrics - Measures of central tendency, dispersion (variance, standard deviation, IQR), and skewness.

- [4.3] Continuous & Discrete Distributions - Normal, Binomial, Poisson, Exponential, and Uniform probability density functions.

- [4.4] Central Limit Theorem & Sampling - Law of large numbers, sampling distribution of the mean, and standard error.

- [4.5] Hypothesis Testing Framework - Null vs Alternative hypotheses, Type I and Type II errors, and significance levels (alpha).

- [4.6] Parametric Tests: t-Tests and ANOVA - One-sample, independent two-sample, paired t-tests, and one-way/two-way ANOVA.

- [4.7] P-Values, Effect Size & Statistical Power - Interpreting p-values, Cohen's d effect sizes, and power analysis.

- [4.8] Correlation vs Causation - Pearson, Spearman rank correlations, confounding variables, and spurious relationships.

- [4.9] Confidence Intervals & Estimation - Point estimates, margin of error, bootstrap resampling, and CI interpretation.

- [4.10] Bayesian Statistical Inference - Prior distributions, likelihood functions, posterior updates, and Markov Chain Monte Carlo (MCMC).

Curriculum Module 05: Time Series Analysis and Predictive Modeling

Scope & Description: Temporal signal decomposition, stationarity, stochastic processes, ARIMA/SARIMA, Holt-Winters, deep forecasting (LSTM), and validation.

Key Sub-Topics & Detailed Syllabus Breakdown:

- [5.1] Introduction to Time Series & Temporal Fundamentals - Continuous timestamps, irregular sampling, trend, seasonality, and residual noise.

- [5.2] Stationarity & Unit Root Tests - Strict vs weak stationarity, differencing, Augmented Dickey-Fuller (ADF), and KPSS tests.

- [5.3] Autocorrelation (ACF) & Partial Autocorrelation (PACF) - Serial correlation, lag analysis, and order identification.

- [5.4] Autoregressive (AR) & Moving Average (MA) Models - White noise assumptions, parameter estimation, and invertibility.

- [5.5] ARIMA & Seasonal ARIMA (SARIMA) - Non-seasonal (p,d,q) and seasonal (P,D,Q)s parameterization and diagnostic checking.

- [5.6] Exponential Smoothing & Holt-Winters - Single, double (trend), and triple (multiplicative/additive seasonal) smoothing algorithms.

- [5.7] Feature Engineering for Temporal Data - Rolling window aggregates, expanding statistics, cyclic calendar features, and lag creation.

- [5.8] Machine Learning for Forecasting - Gradient boosted regression (XGBoost, LightGBM) and Facebook Prophet decomposition.

- [5.9] Deep Learning for Sequential Data - Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU).

- [5.10] Validation & Forecast Evaluation - Time-series cross-validation (rolling origin), MAE, RMSE, MAPE, and production monitoring.

Course Evaluation & Laboratory Integration Guidelines

1. Laboratory Practical Exercises: Each student must complete hands-on Jupyter Notebook experiments mapped directly across Modules 01 to 05.
2. Code Quality & Documentation: All scripts must adhere to PEP-8 standards with reproducible random seeds and explicit markdown documentation.
3. Mid-Term & Final Evaluation: Continuous assessment based on lab reports, live coding challenges, and an end-semester predictive capstone project.
4. Learning Resources: Access interactive web tools, embedded PDF reference manuals, and live code runners on the official Data Science Digital Library platform.

Verified and Approved by:

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